ELSEVIER

A Comparison of Vegetation Indices over a Global Set of TM Images for EOS-MODIS

A. R. Huete, H. Q. Liu,* K. Batchily,* and W. van Leeuwen**

A set of Landsat Thematic Mapper images representing a wide range of vegetation conditions from the NASA Landsat Pathfinder, global land cover test site (GLCTS) initiative were processed to simulate the Moderate Resolution Imaging Spectroradiometer (MODIS), global vegetation index imagery at 250 m pixel size resolution. The sites included boreal forest, temperate coniferous forest, temperate deciduous forest, tropical rainforest, grassland, savanna, and desert biomes. Differences and similarities in sensitivity to vegetation conditions were compared among various spectral vegetation indices (VIs). All VIs showed a qualitative relationship to variations in vegetation. However, there were significant differences among the VIs over desert, grassland, and forested biomes. The normalized difference vegetation index (NDVI) was sensitive to and responded primarily to the highly absorbing red reflectance band, while other indices such as the soil and atmosphere resistant vegetation index (SARVI) were more responsive to variations in the near-infrared (NIR) band. As a result, we found the NDVI to mimic red reflectances and saturate over the forested sites while the SARVI, by contrast, did not saturate and followed variations in NIR reflectances. In the arid and semiarid biomes, the NDVI was much more sensitive to canopy background variations than the SARVI. Maximum differences among vegetation index behavior occurred over the evergreen needleleaf forest sites relative to the deciduous broadleaf forests and drier, grassland, and shrub sites. These differences appear to be useful in complementing the NDVI for improved monitoring of vegetation, with the NDVI sensitive to fraction of absorbed photosynthetic active radiation and the SARVI more sensitive to structural canopy parameters such as leaf area index and leaf morphology. ©*Elsevier Science Inc., 1997*

* Department of Soil, Water, and Environmental Science, University of Arizona, Tucson

Address correspondence to Alfredo R. Huete, Dept. of Soil, Water, and Environmental Science, Univ. of Arizona, 429 Shantz Bldg. #38, Tucson, AZ 85721.

Received 11 December 1995; revised 22 April 1996.

INTRODUCTION

Operational monitoring of the Earth's vegetative cover currently involves the utilization of spectral vegetation indices (VIs) as a precise radiometric measure of the spatial and temporal patterns of vegetation photosynthetic activity. The normalized difference vegetation index (NDVI), which is the difference of the near-infrared (NIR) and red bands divided by their sum, has been the most widely used index in global vegetation studies (Goward et al., 1985; Townshend et al., 1994; Tucker and Sellers, 1986; Malingreau et al., 1989). The NDVI, as measured from the NOAA-Advanced Very High Resolution Radiometer (AVHRR) sensor series, has been successfully utilized in phenological studies of the vegetation growing season, land cover classification, famine early warning detection, and global climate system models (Justice et al., 1985; Sellers et al., 1994; Townshend et al., 1991). These studies have shown a qualitative correspondence of differences in NDVI with observed or recognized variations in vegetation growth.

Progress toward improved monitoring of vegetation patterns and biospheric processes has been partly restricted by the limitations of the AVHRR sensor and partly by the inherent limitations of using empirical or semiempirical spectral indices (Goward et al., 1991; Myneni et al., 1995). Goward et al. (1991) state that errors of ±50% are possible with the NOAA-NDVI product owing to such factors as poor sensor calibration, poor image registration, variable angular-induced pixel sizes, and incomplete cloud screening. Furthermore, the broad visible and near-infrared bands of the AVHRR (Table 1) overlap the sensitive chlorophyll-absorbing "red" region and include several atmospheric water absorption bands.

A major thrust of the Earth Observing System (EOS) program is to move toward a more quantitative

REMOTE SENS. ENVIRON. 59:440–451 (1997)
©Elsevier Science Inc., 1997
655 Avenue of the Americas, New York, NY 10010

0034-4257/97/817.00
PII S0034-4257(96)00112-5

Table 1. Comarison of MODIS Sensor Characteristics with Landsat TM and NOAA–AVHRR

Band/Sensor	*NOAA–AVHRR*	*Landsat TM*	*MODIS*
Blue		0.45–0.52 μm	0.459–0.479 μm
Green	0.55–0.68 μm	0.52–0.60 μm	0.545–0.565 μm
Red		0.63–0.69 μm	0.620–0.670 μm
Near-infrared	0.725–1.1 μm	0.76–0.90 μm	0.841–0.876 μm
Middle-IR			1.230–1.250 μm
Middle-IR		1.55–1.75 μm	1.628–1.652 μm
Middle-IR		2.08–2.35 μm	2.105–2.155 μm
Pixel size at nadir	1 km	30 m	250 m and 500 m
Temporal	Daily	16 days	1–2 days

effort in understanding global biospheric processes. This will require an increased understanding of the global distribution of vegetation types and knowledge of their biophysical properties and temporal variations. The Moderate Resolution Imaging Spectroradiometer (MODIS), onboard the A.M. platform to be launched in 1998, is the primary EOS sensor for providing data on terrestrial biospheric dynamics and process rates (Salomonson et al., 1989; Running et al., 1994). The 36-channel MODIS instrument will include seven shortwave, "land" bands, similar to the Landsat Thematic Mapper (TM), which will provide continuous, 1 to 2-day global coverage at 250 m and 500 m resolutions (Table 1).

A shift from the AVHRR to MODIS sensor for global vegetation studies will involve a change from 1 km (local area coverage, LAC) and 4 km (global area coverage, GAC) pixel resolutions, at nadir, to 250 m red and NIR MODIS bands. The MODIS spectral bandwidths will also be finer and will avoid the water absorption regions in the NIR, such as at 0.82 μm and 0.905 μm (Table 1). Improved algorithm products, such as an operational gaseous and aerosol correction, will further reduce atmospheric contamination for terrestrial-based algorithm products (King et al., 1992; Teillet and Fedosejevs, 1995; Vermote et al., 1994a). Global repetitive coverage of radiometrically calibrated data sets, corrected for atmosphere and with improved spectral and spatial resolutions, will allow for more accurate, large-scale change detection of vegetation conditions, land cover discrimination, and characterization of vegetation biophysical parameters (Townshend et al., 1991).

These improvements will inevitably result in new measures of vegetation, whether they be global, land cover dependent, or vegetation parameter specific. The success and accuracy of the NDVI and newly developed spectral measures for improved global vegetation monitoring, however, will also depend on how well the vegetation algorithms themselves are able to depict actual spatial and temporal vegetation differences. With improved sensor characteristics and atmospheric contamination reduced, we expect to see more information and variability of actual ground surface conditions, including not only green vegetation but also canopy substrate backgrounds and senesced and woody biomass. Thus, enhanced sensor characteristics will improve not only the detection of the vegetation signal but also nonvegetation signals.

Significant progress has also been made in improving the VI equations themselves, primarily in reducing canopy background, atmospheric contamination, and saturation problems in the NDVI (Huete et al., 1994). Currently, two operational vegetation indices are proposed for use with MODIS data. The NDVI will serve as the "continuity" index while an improved VI, more sensitive to biomass variations over a wider range of vegetation conditions and resistant to atmospheric and canopy background effects, will be implemented. There remains interest in atmospherically resistant VIs in order to minimize residual atmospheric variations within an aerosol correction (dark-object approach) grid, which will likely be 50 km resolution (Vermote et al., 1994). Residual artifacts include smoke, gaseous and particulate pollutants, and light cirrus clouds.

The soil adjusted vegetation index (SAVI) was developed to more accurately model NIR radiant transfer in incomplete vegetation canopies in order to remove canopy background noise (Huete, 1988):

$$\mathrm{SAVI}=(1+L)(\rho_{\mathrm{nir}}-\rho_{\mathrm{red}})/(\rho_{\mathrm{nir}}+\rho_{\mathrm{red}}+L), \qquad (1)$$

where the ρ's are reflectances and L is an adjustment factor that accounts for differential red and NIR extinction through the canopy. In a photosynthetically active canopy, red extinction through a canopy will exceed that in the NIR resulting in more NIR energy reflecting off the canopy background relative to the red. This spectrally dependent, secondary signal is not amenable to simple ratioing and a "background correction" (L) becomes necessary.

Kaufman and Tanré (1992) developed the atmospherically resistant vegetation index (ARVI) to minimize atmospheric-induced variations in the VI on a pixel by pixel basis. The ARVI utilizes the difference in radiance between the blue and the red channel, via a γ function, to correct the radiance in the red channel and stabilize

the index to temporal and spatial variations in atmospheric aerosol content:

$$\mathrm{ARVI}=(\rho^*_{\mathrm{nir}}-\rho^*_{rb})/(\rho^*_{\mathrm{nir}}+\rho^*_{rb}), \quad (2)$$

where

$$\rho^*_{rb}=\rho^*_r-\gamma(\rho^*_b-\rho^*_r), \quad (2')$$

and the ρ^*'s are reflectances with prior correction for molecular scattering and ozone absorption. Both atmospheric and canopy background corrections can be combined into the soil and atmospherically resistant vegetation index (SARVI), which would minimize both soil and atmospheric noise and result in a more stable NDVI variant equation (Kaufman and Tanré, 1992):

$$\mathrm{SARVI}=(1+L)(\rho^*_{\mathrm{nir}}-\rho^*_{rb})/(\rho^*_{\mathrm{nir}}+\rho^*_{rb}+L). \quad (3)$$

A further reduction in atmospheric and soil contamination was made by Liu and Huete (1995) by incorporating both soil adjustment and atmospheric resistance concepts into a feedback-based equation, resulting in a modified normalized difference vegetation index (MNDVI), which, expressed in terms of NDVI, is

$$\mathrm{MNDVI}=\mathrm{NDVI}(1+C_2H_2)/(1+C_1H_1), \quad (4)$$

where H_1 and H_2 are functions of the blue, red, and NIR reflectances and atmospheric-, canopy background-, and vegetation-feedback coefficients. The constants C_1 and C_2 are weight functions dependent upon whether the reflectance data were uncorrected, partially corrected, or totally corrected for atmosphere. This equation is simplified by eliminating the requirement that it be expressed in terms of NDVI, resulting in a feedback-based, soil and atmosphere resistant vegetation index (SARVI2) (Huete et al., 1996):

$$\mathrm{SARVI2}=2.5(\rho_{\mathrm{nir}}-\rho_{\mathrm{red}})/(1+\rho_{\mathrm{nir}}+6\rho_{\mathrm{red}}-7.5\rho_{\mathrm{blue}}), \quad (5)$$

where the ρ's are "apparent" directional reflectances (top-of-the-atmosphere). SARVI2 as well as ARVI, SARVI, and MNDVI have been shown to remove smoke plumes and cirrus clouds from Landsat TM imagery, an indication of its ability to minimize residual atmospheric aerosols on a pixel by pixel basis (Huete et al., 1996).

An important part of development and validation of MODIS land products is a network of global test and monitoring sites, selected to be broadly representative of the global range of land covers (Running et al., 1994). In this study, a set of TM images, representing a wide range of vegetation conditions from the U.S. long-term ecological research (LTER) network (Franklin et al., 1990) and the NASA Landsat Pathfinder, global land cover test site (GLCTS) initiative, were processed to simulate MODIS global vegetation index imagery at 250 m pixel size resolution. These data are currently available to the global change research community as part of the NASA Landsat Global Change Data collection at the EROS Data Center, Sioux Falls. As part of an overall, global validation effort, we compare the capability and performance of MODIS-like vegetation index products in depicting spatial and temporal variations in vegetation from deserts and grasslands to woodlands, temperate forests, and tropical rainforests. We also assess the functional equivalence of various equations to determine if there is additional information present in some of the indices or if they are depicting the same information within and between land cover types (Perry and Lautenschlager, 1984).

METHODS

A set of 16 Landsat 4 and 5 Thematic Mapper images were processed into MODIS vegetation index images. The 30 m pixel resolution images were first radiometrically calibrated and converted into top-of-the-atmosphere, "apparent" reflectances using exoatmospheric irradiance inputs and an updated set of digital count to radiance coefficients from TM satellite-vicarious calibration experiments (Thome et al., 1993). A Rayleigh and ozone correction were then applied using the "6S" radiative transfer code (Vermote et al., 1994b). The TM images were transformed into 250 m, MODIS-simulated images with a Fourier transformation program. The single-channel reflectance images were then utilized in computation of NDVI, SAVI, ARVI, SARVI, and SARVI2 images. Data were extracted in 4×4 "windows" over uniform areas, representative of the dominant land cover types. From 8 to 10 windows were extracted for each major land cover type.

All of the TM images were 1992 acquisitions between the months of July and September (Table 2). The LTER sites included eastern deciduous broadleaf forests (Harvard Forest and Hubbard Brook), evergreen needleleaf forests (H. J. Andrews Experimental Forest), mixed evergreen and deciduous forests (North Temperate Lakes), tallgrass (Konza Prairie) and shortgrass (Central Plains Experimental Range) prairies, semiarid mixed grass-shrub (Sevilleta National Wildlife Refuge), and desert shrub (Jornada Experimental Range). Another set of TM images included four African National Parks, representing wet and dry savanna (Kasanka and Kruger National Parks, respectively), edaphic grassland (Kafue National Park), and xeromorphic shrubland (Etosha National Park). A fifth African image (also dry savanna) was from the Hydrologic Atmospheric Pilot Experiment in Sahel (HAPEX-Sahel), centered at the "southern supersite" at the peak of the growing season (Prince et al., 1995). The remaining TM images included a boreal forest site in Siberia and tropical rainforest sites in the Amazon Basin (Rondonia and Maraba) (Nelson, 1994).

RESULTS

Deserts, Savannas, and Grasslands

We first analyzed the nonforested sites covered in the TM imagery, including playa lakebeds devoid of vegeta-

Table 2. Vegetation Type and Locations of the Landsat TM Images

	Site Name	*Long.*	*Lat.*	*Solar Angle, Zenith*	*Date (TM 4/5)*	*Precip./Temp.*
1	Hubbard Brook LTER: deciduous forest (northern hardwood and spruce-fir)	71 W	44 N	31	2 Jul 92 (5)	120 cm/5°C
2	Harvard Forest LTER: deciduous forest (hemlock-white pine)	72.2 W	42.4 N	30	2 Jul 92 (5)	110 cm/8°C
3	North Temperate Lakes LTER: mixed deciduous-conifer forest	89 W	46 N	45	8 Sep 92 (5)	75 cm/5°C
4	H. J. Andrews LTER coniferous forests (Western Hemlock, Ponderosa pine)	122 W	44 N	36	10 Aug 92 (5)	250 cm/10°C
5	Konza Prairie LTER tallgrass prairie	96.6 W	39.1 N	37	26 Aug 91 (5)	82 cm/13°C
6	Central Plains LTER shortgrass prairie	104.7 W	40.8 N	37	15 Aug 92 (5)	32 cm/9°C
7	Sevilleta, N.M. LTER shortgrass-desert transition	105 W	34 N	35	15 Aug 92 (5)	25 cm/22°C
8	Jornada LTER Desert (desert grass-shrub)	106.7 W	33.1 N	30	6 Jul 92 (4)	22.5 cm/20°C
9	Kasanka National Park, Zambia [subtropical, wet (Miombo) savanna]	30.2 E	12.5 S	34	14 Aug 92 (4)	120 cm
10	Kruger National Park, South Africa, Miombo woodland	31.5 E	24.5 S	42	16 Aug 92 (4)	60 cm
11	Kafue National Park, Zambia edaphic grassland, Miombo woodland	26 E	15 S	40	27 Jul 92 (4)	81 cm/22°C
12	Etosha National Park, Namibia Xeromorphic shrubland	14 E	18 S	39	28 Jul 92 (4)	56 cm/30°C
13	HAPEX-Sahel	2.5 E	13.5 N	34	25 Sep 92 (4)	60 cm/29°C
14	Siberia Boreal forest	129 E	61 N	46	8 Aug 92 (5)	24.5 cm/−8°C
15	Rondonia, Ji Parana Open tropical–savanna transition	61.0 W	10.7 S	49	10 Jul 92 (4)	211 cm/25°C
16	Maraba, Eastern Amazon dense, humid tropical forest	49.2 W	5.5 S	41	10 Aug 92 (4)	205 cm/25°C

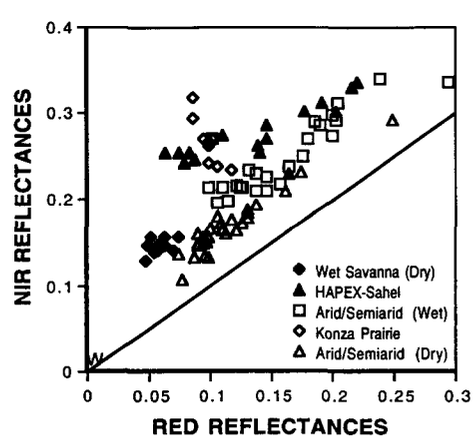
tion, arid/semiarid areas measured in both the dry and wet seasons, wet savanna, and short and tall grasslands. Combined, these sites span the range from zero to dense amounts of actively photosynthetic vegetation. The "dry season" sites were mostly in Africa, south of the equator, and included Etosha National Park, Kruger National Park, and Kafue National Park. The "greener," wet season sites with more photosynthetic activity included Jornada, a "hot desert," Sevilleta, a mixed grass-shrub site, Central Plains, a shortgrass prairie, and Konza Prairie, a tall grassland. The HAPEX-Sahel site, measured during the rainy season, included a gradient of vegetation transitions with increasing green biomass and rainfall from the northern to the southern portion of the image (Prince et al., 1995). Kasanka National Park, although measured during the dry season, encompassed a more humid, wet savanna with green vegetation noticeably present. The Konza tallgrass prairie contained the densest cover of photosynthetically active vegetation. A set of pixels from a lake at the North Temperate Lakes, LTER site was also extracted for reference purposes.

A red–NIR scatterplot (Fig. 1) shows the dry, arid, and semiarid sites closest to the 1:1 diagonal line and the Konza tallgrass prairie and more humid portions of HAPEX furthest from the 1:1 line. The arid/semiarid sites during their wet season were at an intermediate distance. Primary variations, however, were scattered along the 1:1 diagonal line, rather than away from it. This was due to spectral variations in arid and semiarid regions being more associated with variations in geologic substrate materials (soil, rocks, litter) rendering canopy background variations most dominant.

In Figure 2 the general relationships of the NDVI and SARVI2 are plotted against the red and NIR reflectances to assess differences in VI behavior. There is a strong negative trend in the NDVI vs. red reflectance plot (Fig. 2a). Two distinct trends were clearly recognizable with a lower (left) curve depicting the "dry season" sites and the upper (right) curve depicting the "wet/season", subhumid and semiarid sites. In all cases, the darker the site (lower red reflectance), the higher the NDVI, regardless of whether the lower red reflectance was due to the presence of green biomass or to variations in substrate brightness.

SARVI2, on the other hand, showed VI behavior nearly independent of red reflectances over the "dry season" arid/semiarid sites (Fig. 2b). Over the entire "dry" zones of Etosha, Kruger, and Kafue, the NDVI varied by 0.20 units (0.05–0.25) while the SARVI2 values varied by 0.08 units (0.05–0.13). At Etosha National Park, a xeromorphic, sparse shrubland with almost no photosynthetic activity, the NDVI varied by 0.10 NDVI units, increasing with decreasing surface brightness while SARVI2, being more independent of surface brightness, varied by only 0.02 units over this area. Similar results

Figure 1. Red and NIR reflectance cross-plot of the desert to tall grass land cover types (W is a deep lake, water reference).



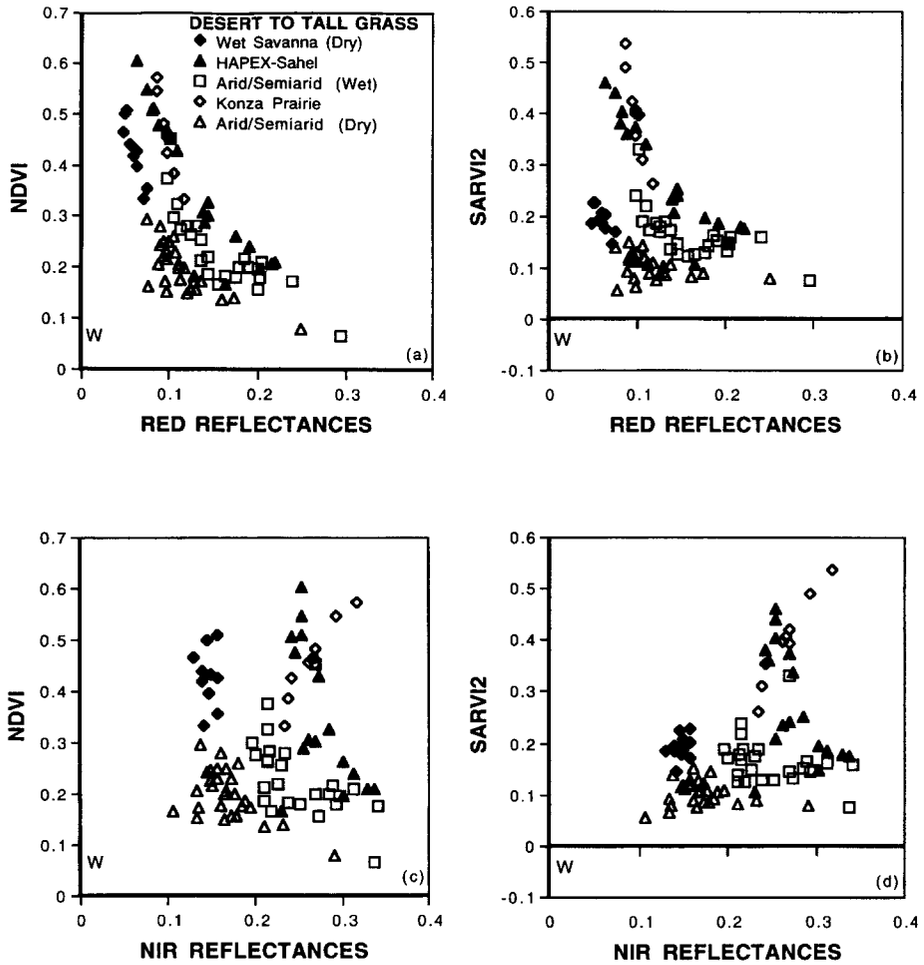


Figure 2. NDVI and SARVI2 relationships with the individual red and NIR reflectances of the desert to tall grass land cover types.

in the arid zones of the Sahara region were reported by Huete and Tucker (1991). SARVI2 over the subhumid sites and "wet season" semiarid sites behaved similarly to the NDVI, a result of not only increasing green biomass, but also less soil contributions and less variability in soil brightness. Crossplots of the two VIs with NIR reflectances showed less clear relationships (Figs. 2c,d). There is still a canopy background influence in the NDVI where lower NIR reflectances resulted in higher NDVI values in the "dry season," arid and semiarid, and HAPEX sites (Fig. 2c). The SARVI2-NIR crossplot shows the "dry season" and arid/semiarid sites to have SARVI2 values nearly independent of NIR reflectances and the subhumid sites (Konza and HAPEX) to be positively related with NIR reflectances (Fig. 2d).

The biggest difference between the two VIs occurred over the Kasanka site (Savanna) which had very high NDVI values but relatively low SARVI2 values (Fig. 2). The NDVI shows this site to be as densely vegetated as the Konza Prairie and humid portions of HAPEX, while the SARVI2 values show the "dry season" Kasanka site to be similar to the semiarid sites during the wet season. The red–NIR scatterplot (Fig. 1) showed the Kasanka site to have among the lowest red and NIR reflectances, plotting not too far from the diagonal line. The high NDVI value of the Kasanka site may either be due to its dark signature or to the presence of dense green biomass. One must also leave open the possibility that the two VIs may be responding to different structural canopy parameters. A comprehensive "ground-truth" validation would be required for a definitive explanation.

Further insights into the nature of the VIs can also be made by comparing their spectral reflectance signatures relative to the VI values. Figure 3a shows the "dry season" spectral signatures of the African sites. These signatures show little evidence of the presence of "green" vegetation, particularly between the red and NIR bands. The NDVI values, nevertheless increased threefold (0.08–0.22) as the spectral signature became darker, while the SARVI2, which minimizes canopy substrate variations, showed very little difference in values (0.08–0.11) (Fig. 3a). The same can be seen in a plot of "wet season," semiarid spectral signatures (Fig. 3b), where the NDVI doubled in value (0.20–0.44) as the signatures became darker, while the SARVI2 varied slightly (0.15–0.21). In both of the previous plots, there were no significant changes in the red to NIR portions of the spectral signatures to warrant large changes in the VI val-

ues. Note also the Kasanka site signatures with much higher NDVI values but only slight differences in red-to-NIR spectral signatures.

In Figure 3c, the spectral signatures of the subhumid sites (Konza and HAPEX) show the much steeper red–NIR signatures corresponding to the higher NDVI and SARVI2 values. There is greater variation in the NIR and SARVI2 values than there is in the visible and NDVI values. For comparison, the Kasanka site signatures are also plotted. Once again, their spectral signatures do not seem to justify such high amounts of green vegetation as indicated by the high NDVI values.

Forested Areas

In this section, we report on the analyses of VI responses over the forested sites, which included tropical rainforests (Rondonia and Maraba) measured in the dry season, temperate deciduous broadleaf forests (Harvard Forest and Hubbard Brook), mixed deciduous-conifer stands (North Temperate Lakes), boreal forest (Siberia), and temperate needleleaf forests (Coastal Range, Cascade Mountains, and the forested "Sky Islands" of the desert southwest).

The forested sites showed very little variation in red reflectances (0.02–0.07) and much larger variations in NIR reflectances (0.16–0.50) (Fig. 4). The two VIs had varying sensitivities to the individual band reflectances with the NDVI showing a strong, negative correlation with red reflectances while the SARVI2 displayed a strong, positive correlation with NIR reflectances. The NDVI had a weaker correlation with the NIR and became saturated over many sites at approximately 0.85. The SARVI2, on the other hand, had essentially no correlation with the red reflectances. The NDVI ranged from 0.55 to 0.90 over the forested sites, while the SARVI2 varied from 0.30 to 0.90. The NDVI appeared to respond more to the red reflectances and canopy parameters associated with absorbed photosynthetically active radiation, whereas the SARVI2 was much more responsive to the NIR reflectances and associated structural canopy parameters such as leaf area index (LAI).

An analysis of the forest spectral signatures (Fig. 5) also confirms this result. One can see very slight variations in the visible bands with large variations in the NIR reflectances over the deciduous and coniferous forests (Figs. 5a,b). The top four signatures, representing deciduous forest sites, are nearly identical in NDVI values yet vary by nearly 0.30 SARVI2 units (Fig. 5a). The lowest two signatures are from the mixed deciduous-conifer stands and decrease in both VI values and NIR reflectances. A similar pattern is observed with the evergreen needleleaf forests (Fig. 5b), which had very high NIR variations (0.15–0.45) and low red reflectance variations. The NDVI and SARVI2 values varied by 0.25 and 0.55 units, respectively. Red reflectances were slightly more

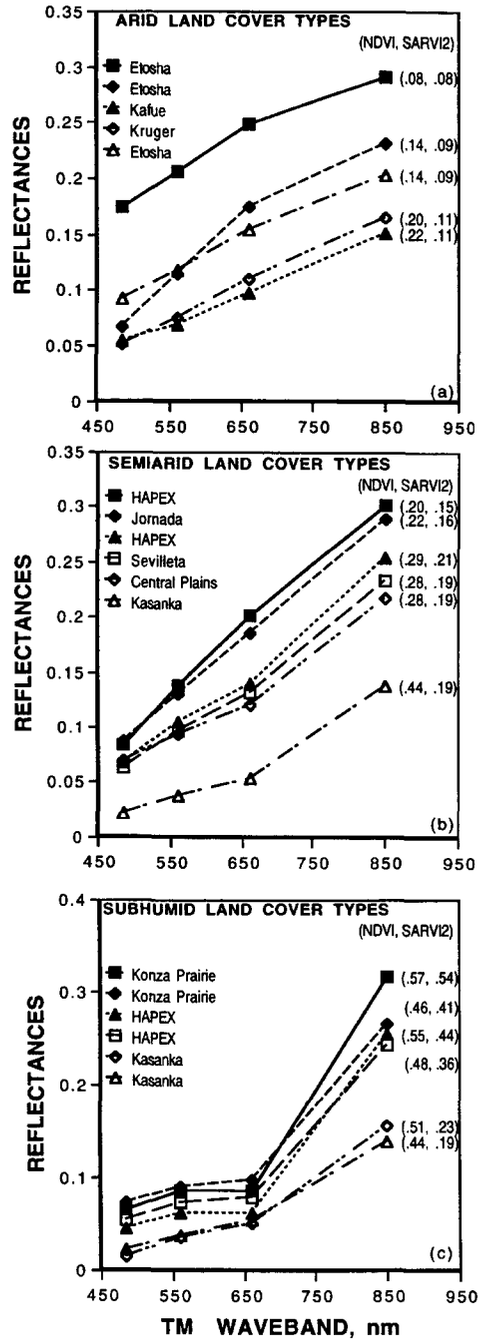


Figure 3. Spectral reflectance signatures for representative a) arid, b) semiarid, and c) subhumid land cover types.

variable than in the case of the deciduous broadleaf forests due to the needle-leaf structure of coniferous forests, which project less leaf area than the broadleaf forests with "flatter," more exposed leaves.

In Figure 6a, a series of spectral reflectance signatures were extracted along the Oregon Transect. The humid Coastal Range (250 cm precip.) typically produces LAI values of around 7, whereas the drier Cascade

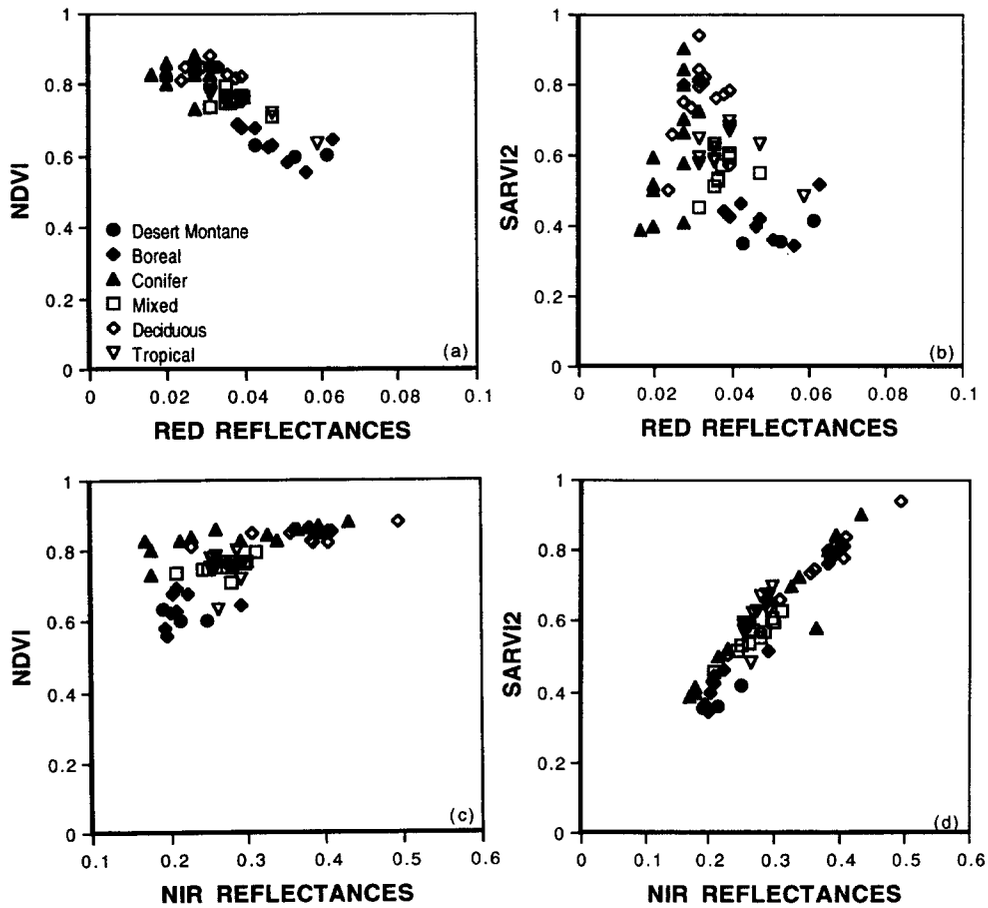


Figure 4. NDVI and SARVI2 relationships with the individual red and NIR reflectances of the forested land cover types.

Mountains (180 cm precip.) consists of less densely vegetated Ponderosa Pine and Western Hemlock forests with LAIs of 2–3 (Runyon et al., 1994; Spanner et al., 1994). The NDVI, with values between 0.80 and 0.88, was not very sensitive to the large range in LAI within these forested sites, while the SARVI2 varied considerably (0.40–0.90). Runyon et al. (1994), in a study of LAI–NDVI relationships across the Oregon Transect, experiment found saturation to occur with NDVI values of around 0.75. A crossplot of the two VIs with NIR reflectances (Fig. 6b) further shows the insensitivity of the NDVI to NIR variations in densely vegetated forests, and hence its inability to depict LAI variations.

In Figure 7, a global range of forested spectral signatures are shown, including deciduous, evergreen, and tropical forests. The greatest variation in reflectances occur in the NIR, and thus VIs that respond more to the NIR than to the visible reflectances will similarly display large variations. The SARVI2 varied by 0.60 units compared with 0.25 units for the NDVI. The tropical rainforest sites produced lower NIR reflectances than many of the temperate forests. This may be due to these sites being measured in the dry season, just before the onset of the rains. Also, the more humid tropical atmospheres may have affected the NIR signal more so than in the temperate forests since the TM data set was not corrected for water vapor within the TM filters.

Global Analysis

In this section, we include all the vegetation types in the TM data set from deserts to forests. In Figure 8a, the red-NIR scatterplot shows the arid and semiarid sites to fall mostly along or close to the diagonal, 1:1 line. The forested sites are concentrated on the left-hand side, along the NIR axis, with nearly constant red reflectances, but varying NIR reflectances. One can see the much darker appearances of the conifer forests relative to the deciduous forests. A scatterplot of the two VIs over all vegetation types shows the dynamic range from "zero" vegetation to dense forest to be nearly the same for the two VIs (Fig. 8b). The NDVI values are always higher than those of the SARVI2, particularly over the "dry season" sites in Africa and over the evergreen needleleaf forests. Maximum differences between the two VIs were highest at intermediate values (~0.5), with less differences over the deciduous broadleaf forests and the arid/

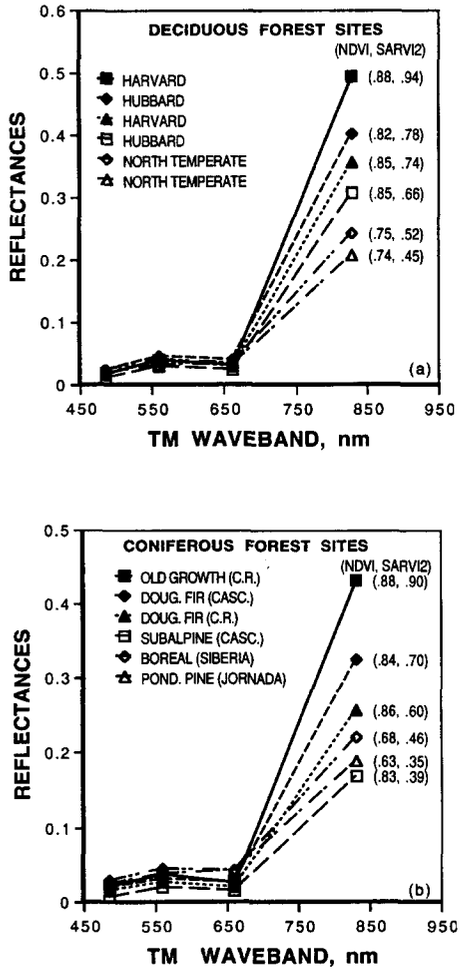


Figure 5. Spectral reflectance signatures for representative a) deciduous forest and b) coniferous forest sites.

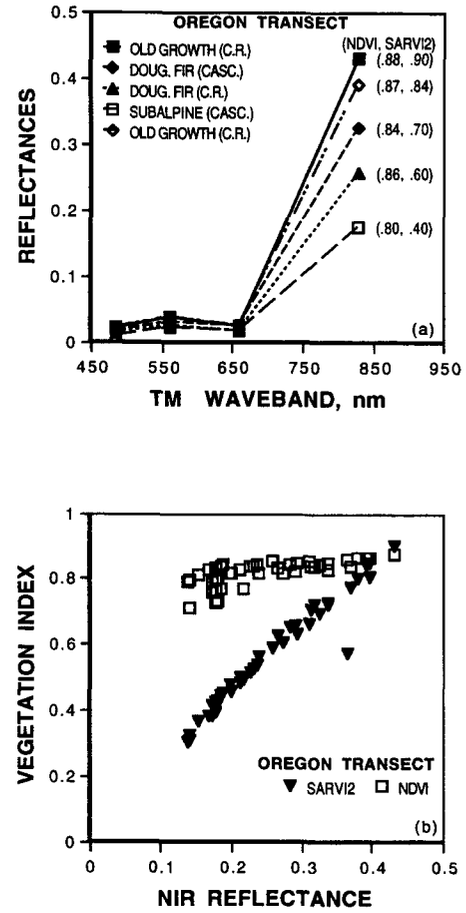


Figure 6. Spectral reflectance signatures (a) and VI–NIR relationships (b) for conifer forest sites along the Oregon Transect.

semiarid deserts and grasslands (Fig. 8b). The saturation of the NDVI is also readily seen, and it appears that the SARVI2 has not saturated at all.

In Figure 9, a summary of the relationships of the two VIs with red and NIR reflectances is shown. The NDVI is most strongly coupled to the red reflectances (Fig. 9a) while the SARVI2 is strongly coupled with the NIR reflectances (Fig. 9d). The NDVI is not well correlated with NIR reflectances, and the saturation of the NDVI at high levels of green biomass is apparent in the top portion of Figure 9c, where a large range of NIR values yield nearly constant NDVI values. The SARVI2 is still negatively correlated with red reflectances (Fig. 9b).

The other vegetation indices analyzed in this study included the ARVI, SAVI, and SARVI. The ARVI behaved similarly to the NDVI (r=0.96), and the SAVI and SARVI behaved similar to the SARVI2 (r=0.99 for both) (Fig. 10).

Thus the SARVI2, SARVI, and SAVI all were highly responsive to vegetation canopy NIR spectral variations and did not saturate over the densely vegetated forests. Similarly, the ARVI was more sensitive to "red reflectances" and saturated quickly as with the NDVI. Canopy background influences in the ARVI were also prevalent and similar to those encountered with the NDVI. This is somewhat expected since the ARVI is an atmospherically resistant version of the NDVI; and the SARVI2 and SARVI are atmospherically resistant versions of the SAVI. These indices incorporate a blue band to reduce aerosol contamination without altering the fundamental behavior and interpretation of an index to surface vegetation.

DISCUSSION

Global change research mandates the creation and maintenance of long-term, high-quality data sets while instrument technologies and new algorithm equations evolve. In this study, Landsat TM imagery were used to better understand the characteristic changes in VIs with a shift from the broad-band, AVHRR to the much more narrow-band, MODIS sensor. The robustness of several VIs

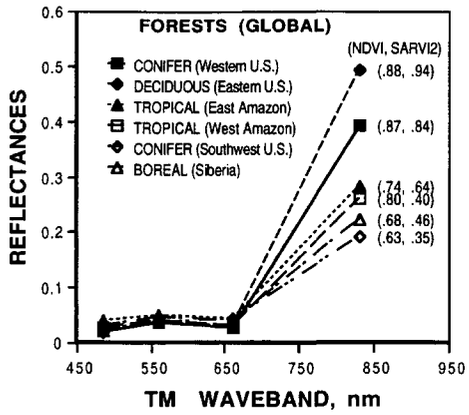


Figure 7. Spectral reflectance signatures for a global range of forest types.

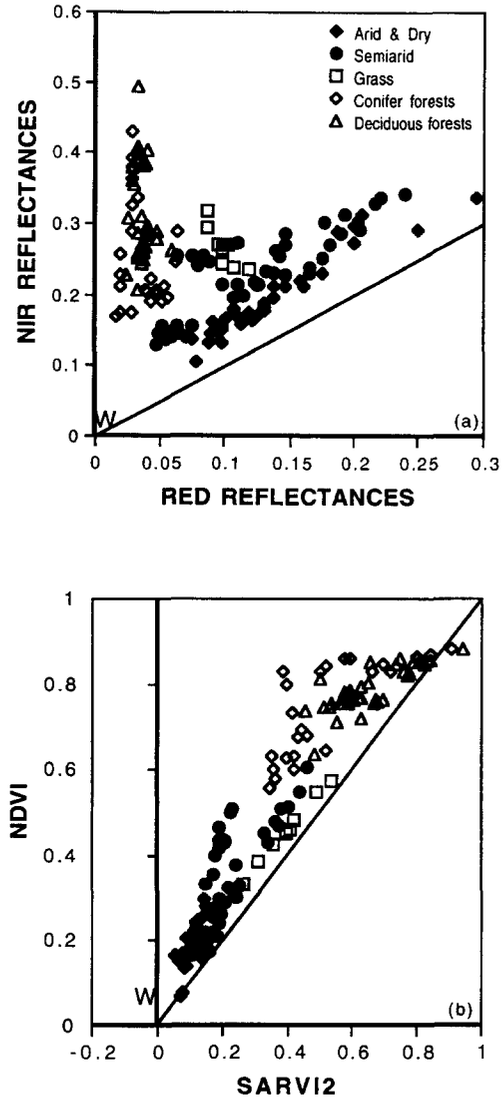


Figure 8. Red and NIR reflectance crossplot (a) and NDVI and SARVI2 crossplot (b) of the global range of land cover types.

over a global range of vegetation conditions was demonstrated in this study. All of the indices exhibited a healthy dynamic range from arid regions to densely vegetated forests. There were significant, secondary differences, however, among the VIs, which suggests that an additional VI may complement the NDVI in global vegetation monitoring studies in the Earth Observation System (EOS) era. The results of this study indicate that the SARVI2, SARVI, and SAVI can refine global vegetation studies by extending the range of sensitivity of VIs into both densely vegetated, forested regions, and sparsely vegetated, arid and semiarid regions.

Increased sensitivity in densely vegetated areas was accomplished by placing more weight to the NIR reflectance component of the SARVI2, SARVI, and SAVI equations. The less absorbing and greater penetration and scattering properties of the NIR enabled increased sensitivity to green biomass and prevented VI saturation at the highest levels of vegetation encountered in this study. The NDVI was shown here, as in other studies (Gitelson et al., 1996), to "saturate" over densely vegetated areas, whereby the NDVI no longer responded to variations in green biomass. This was attributed to the high sensitivity of the NDVI to the red (chlorophyll) absorption band, which also saturated quickly. There is concern of increased saturation problems in MODIS-NDVI imagery resulting from the "finer" and more chlorophyll-sensitive red channel (Gitelson et al., 1996). Changes in land use, land cover, and net primary production may be difficult to detect in a "saturated" mode (Townshend et al., 1991). Land cover classification based on multitemporal NDVI profiles would similarly be hampered. An index, such as the SARVI2, with an extended, linear response over a wider range of vegetation conditions would not only minimize the "saturation" problem, but would also allow for more accurate aggregation and scaling of multiresolution data sets.

Increased sensitivity in arid/semiarid regions was accomplished by minimizing canopy background variations, allowing greater discrimination capability between the weak vegetation signal and dominant canopy background signal. We found the NDVI to vary appreciably over arid and semiarid regions as a result of canopy background variations. This restricts the accurate assessment of the vegetative cover, especially since soil spectral variations are far greater in arid/semiarid regions than in the more humid, organic-enriched grasslands and forests. Canopy background considerations are not only important in sparsely vegetated areas but in woodlands, savannas, and open forest stands as well.

The greatest differences among the VIs occurred over the intermediate range of VI values, particularly over the evergreen needleleaf forests. With the exception of the Coastal Range evergreen rainforest, SARVI2 values were much lower over the evergreen needleleaf for-

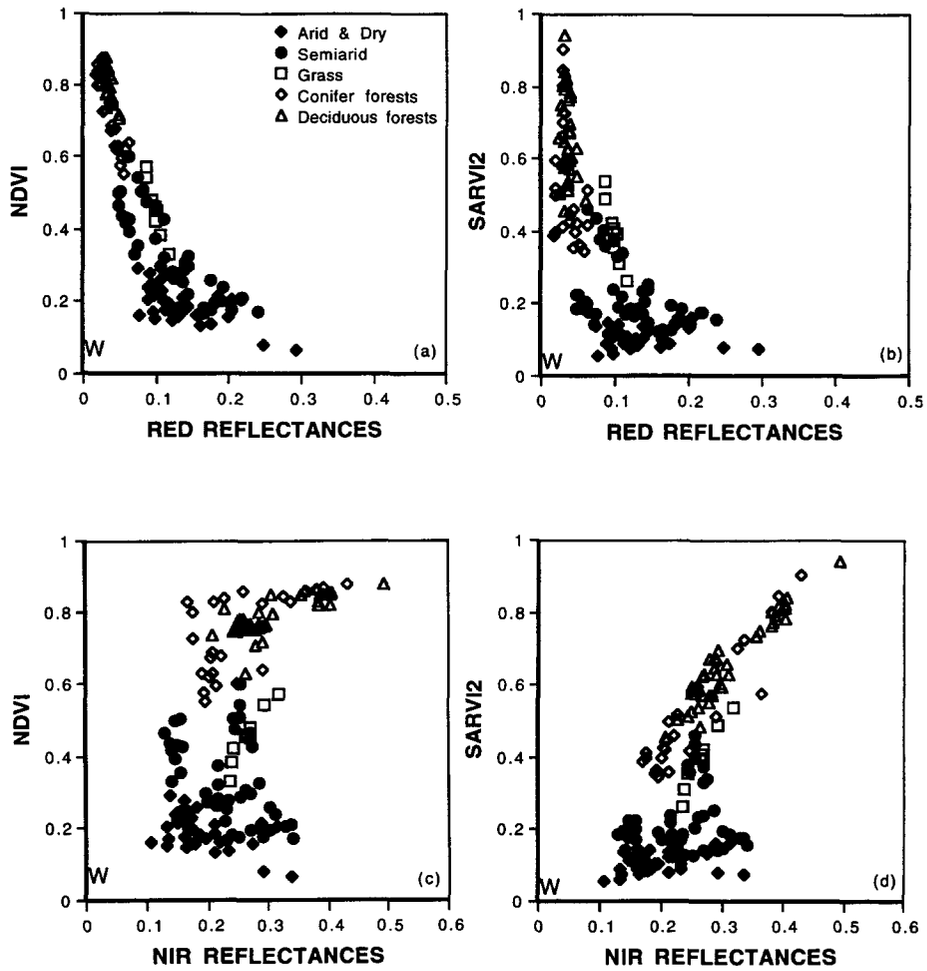


Figure 9. NDVI and SARVI2 relationships with the individual red and NIR reflectances for the global range of land cover types (symbols as in Fig. 8).

ests than broadleaf deciduous forests. The lower SARVI2 values were attributable to the very low NIR reflectances typically encountered in evergreen needleleaf forests, and the higher dependency of the SARVI2 to the NIR channel. The low NIR reflectances, in turn, are associated with the lower extent of projected leaf area in needleleaf forests relative to the more flat and exposed, broadleaf forests. Broadleaf forests have more exposed leaf area for a given leaf biomass and appear much "greener" than needleleaf forests with less projected leaf area.

It is apparent that much of the differences encountered between the NDVI and SARVI2 are a result of their unique sensitivities to the red and NIR reflectances. Epiphanio and Huete (1995) showed differences between the SAVI and NDVI over alfalfa plots with the NDVI varying primarily with red reflectances, while the SAVI was more responsive to NIR reflectances. These two bands depict different information about a vegetation canopy. The reflectance of canopies in the visible part of the spectrum is controlled by the photosynthetic capacity of the canopy to absorb (total chlorophyll absorption), particularly in the red, sensitive part of the visible spectrum (Curran, 1982). In contrast, there is very little absorption of energy in the near-infrared wavelengths, resulting in very high reflectances from not only the uppermost layers, but also from deep within the canopy.

The NDVI was most sensitive to red reflectance variations and appears to be useful in fPAR and percent green cover studies. The NDVI showed less variation in the densely forested areas since there were only minor differences in the photosynthetically active, visible reflectances. The SARVI2, on the other hand, was more sensitive to NIR reflectance variations which are more useful for structural canopy parameters such as LAI, biomass, and leaf morphology. Both NIR reflectances and corresponding SARVI2 values varied considerably within forested canopies and between deciduous broadleaf and evergreen needleleaf morphologies. Bartlett et al. (1988), in a study of divergent canopy morphologies over the same species of cordgrass, found the visible reflectances to remain identical but NIR reflectances differed considerably. The green leaf area per biomass of leaf tissue varied widely and the flatter leaves produced the highest NIR reflectances.

A more sophisticated use of the VIs is to derive canopy biophysical parameters such as fAPAR, LAI, green

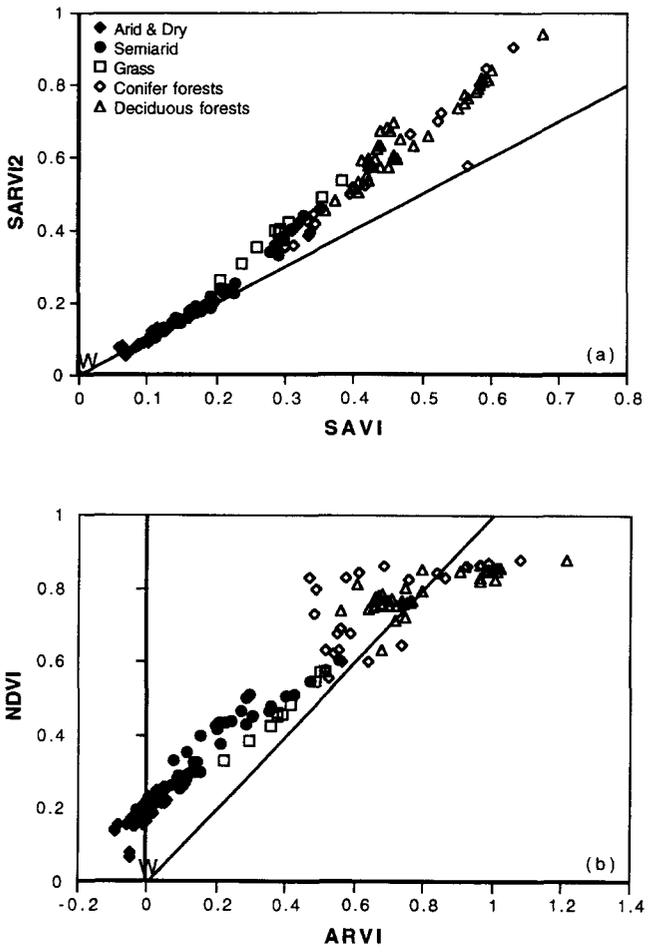


Figure 10. Crossplots of a) SARVI2 and SAVI and b) NDVI and ARVI for the global range of land cover types (symbols as in Fig. 8).

biomass, and percentage green cover (Baret and Guyot, 1991; Asrar et al., 1984; 1992; Goward and Huemmrich, 1992; Sellers, 1985; Sellers et al., 1994). Since, to a certain extent, these four biophysical parameters covary with each other, it is common to find general VI correlations with all of these variables. Many studies have concluded, however, that VI translation is parameter-specific and land-cover-dependent, requiring knowledge of various canopy structural parameters (Running et al., 1994). Bartlett et al. (1988) found separate LAI–biomass and spectral relationships in canopies with varying canopy morphologies. A more quantitative VI to LAI translation will thus require a land cover or biome map with key structural parameters (leaf form, shape, angle). Although a biophysical validation of VIs was not the focus of this study, it is apparent that different VIs are more suited for derivation of specific biophysical vegetation parameters.

The TM imagery utilized here only involved nadir-view VIs and did not consider bidirectional reflectance factor (BRF) issues for "global" VI robustness. Angular considerations of sensor view and solar illumination effects on the VIs must also be resolved, hopefully with BRF corrections to nadir viewing angles. Ratioing of individual bands does not remove surface anisotropy as shown by the forward scattering bias of the NDVI (Gutman, 1991). Similarly, simple ratioing can only reduce topographic variations when surfaces are near-Lambertian in behavior. Surface anisotropy will become more pronounced in the EOS era, as a result of improved atmospheric removal algorithms, which will accentuate differences in bidirectional surface reflectances resulting from canopy structural influences (Cihlar et al., 1994).

REFERENCES

Asrar, G., Fuchs, M., Kanemasu, E.T., and Hatfield, J.L. (1984), Estimating absorbed photosynthetic radiation and leaf area index from spectral reflectance in wheat. *Agron. J.* 76:300–306.

Asrar, G., Myneni, R. B., and Choudhury, B. J. (1992), Spatial heterogeneity in vegetation canopies and remote sensing of absorbed photosynthetically absorbed radiation: a modeling study. *Remote Sens. Environ.* 41:85–101.

Baret, F., and Guyot, G. (1991), Potentials and limits of vegetation indices for LAI and APAR assessment. *Remote Sens. Environ.* 35:161–173.

Bartlett, D. S., Hardisky, M. A., Johnson, R. W., Gross, M. F., Klemas, V., and Hartman, J. M. (1988), Continental scale variability in vegetation reflectance and its relationship to canopy morphology. *Int. J. Remote Sens.* 9:1223–1241.

Cihlar, J., Manak, D., and Voisin, N. (1994), AVHRR bidirectional reflectance effects and compositing. *Remote Sens. Environ.* 48:77–88.

Curran, P. J. (1982), Multispectral remote sensing for estimating biomass and productivity, in *Plants and the Daylight Spectrum* (H. Smith, Ed.), Academic Press, London.

Epiphanio, J. C. N., and Huete, A. R. (1995), Dependence of NDVI and SAVI on sun/sensor geometry and its effect on fPAR relationships in Alfalfa. *Remote Sens. Environ.* 51: 351–360.

Franklin, J. F., Bledsoe, C. S., and Callahan, J. T. (1990), Contributions of the long-term ecological research program. *BioScience* 40:509–523.

Gitelson, A. A., Kaufman, Y. J., and Merzlyak, M. N. (1997), Use of a green channel in remote sensing of global vegetation from EOS-MODIS. *Remote Sens. Environ.* 58:289–298.

Goward S. N., and Huemmrich, K. F. (1992), Vegetation canopy PAR absorptance and the normalized difference vegetation index: an assessment using the SAIL model. *Remote Sens. Environ.* 39:119–140.

Goward S. N., Tucker, C. J., and Dye, D. G. (1985), North American vegetation patterns observed with the NOAA-7 Advanced Very High Resolution Radiometer. *Vegetatio* 64:3–14.

Goward S. N., Markham, B., Dye, D. G., Dulaney, W., and Yang, J. (1991), Normalized difference vegetation index measurements from the Advanced Very High Resolution Radiometer. *Remote Sens. Environ.* 35:257–277.

Gutman, G. (1991) Vegetation indices from AVHRR: an update and future prospects. *Remote Sens. Environ.* 35:121–136.

Huete, A. R. (1988), A soil adjusted vegetation index (SAVI). *Remote Sens. Environ.* 25:295–309.

Huete, A. R., Justice, C., and van Leeuwen, W. (1996), MODIS Vegetation index (MOD 13), EOS MODIS Algorithm—Theoretical Basis Document, NASA Goddard Space Flight Center, Greenbelt, Maryland 20771, USA, 115 pp.

Huete, A. R., Justice, C., and Liu, H. (1994), Development of vegetation and soil indices for MODIS-EOS. *Remote Sens. Environ.* 49:224–234.

Huete, A. R., and Tucker, C. J. (1991), Investigation of soil influences in AVHRR red and near-infrared vegetation index imagery. *Int. J. Remote Sens.* 12:1223–1242.

Justice, C. O., Townshend, J. R. G., Holben, B. N., and Tucker, C. J. (1985), Analysis of the phenology of global vegetation using meteorological satellite data. *Int. J. Remote Sens.* 6:1271–1318.

Kaufman, Y. J., and Tanré, D. (1992), Atmospherically resistant vegetation index (ARVI) for EOS-MODIS. *IEEE Trans. Geosci. Remote Sens.* 30:261–270.

King, M. D., Kaufman, Y. J., Menzel, W. P., and Tanré, D. (1992), Remote sensing of cloud, aerosol, and water vapor properties from the Moderate Resolution Imaging Spectrometer (MODIS). *IEEE Trans. Geosci. Remote Sens.* 30: 2–27.

Liu, H. Q., and Huete, A. R. (1995), A feedback based modification of the NDVI to minimize canopy background and atmospheric noise. *IEEE Trans. Geosci. Remote Sens.* 33: 457–465.

Malingreau, J. P., Tucker, C. J., and Laporte, N. (1989), AVHRR for monitoring global tropical deforestation. *Int. J. Remote Sens.* 10:855–867.

Myneni, R. B., Hall, F. G., Sellers, P. J., and Marshak, A. L. (1995), The interpretation of spectral vegetation indices. *IEEE Trans. Geosci. Remote Sens.* 33:481–486.

Nelson, B. W. (1994), Natural forest disturbance and change in the Brazilian Amazon. *Remote Sens. Rev.* 10:105–125.

Perry, C. R., Jr., and Lautenschlager, L. F. (1984), Functional equivalence of spectral vegetation indices. *Remote Sens. Environ.* 14:169–182.

Prince, S. D., Kerr, Y. H., Goutorbe, J. P., et al. (1995), Geographical, biological and remote sensing aspects of the Hydrologic Atmospheric Pilot Experiment in the Sahel (HAPEX-Sahel). *Remote Sens. Environ.* 51:215–234.

Running, S. W., Justice, C.O., Salomonson, V., et al. (1994), Terrestrial remote sensing science and algorithms planned for EOS/MODIS. *Int. J. Remote Sens.* 17:3587–3620.

Runyon, J., Waring, R. H., Goward, S. N., and Welles, J. M. (1994), Environmental limits on net primary production and light-use efficiency across the Oregon transect. *Ecol. Appl.* 4:226–237.

Salomonson, V. V., Barnes, W. L., Maymon, P. W., Montgomery, H. E., and Ostrow, H. (1989), MODIS: advanced facility instrument for studies of the earth as a system. *IEEE Trans. Geosci. Remote Sens.* 27:145–153.

Sellers, P. J. (1985). Canopy reflectance, photosynthesis and transpiration. *Int. J. Remote Sens.* 6:1335–1372.

Sellers, P. J., Tucker, C. J., Collatz, G. J., et al. (1994), A global 10∘10 NDVI data set for climate studies. Part 2—the adjustment of the NDVI and generation of global fields of terrestrial biophysical parameters. *Int. J. Remote Sens.* 15: 3519–3545.

Spanner, M., Johnson, L., Miller, J., et al. (1994), Remote sensing of seasonal leaf area index across the Oregon transect. *Ecol. Appl.* 4:258–271.

Teillet, P. M., and Fedosejevs, G. (1995), On the dark target approach to atmospheric correction of remotely sensed data. *Can. J. Remote Sens.* 21:374–387

Thome, K. J., Gellman, D. I., Parada, R. J., Biggar, S. F., Slater, P. N., and Moran, M. S. (1993), In-flight radiometric calibration of Landsat-5 Thematic Mapper from 1984 to present, in *Proceedings of SPIE Conference 1938*, Orlando, Florida, pp. 126–131.

Townshend, J. R. G., Justice, C., Li, W., Gurney, C., and McManus, J. (1991), Global land cover classification by remote sensing: present capabilities and future possibilities. *Remote Sens. Environ.* 35:243–256.

Townshend, J. R. G., Justice, C. O., Skole, D., et al. (1994), The 1 km resolution global data set: needs of the International Geosphere Biosphere Programme. *Int. J. Remote Sens.* 15:3417–3441.

Tucker, C. J., and Sellers, P. J. (1986), Satellite remote sensing of primary productivity. *Int. J. Remote Sens.* 7:1395–1416.

Vermote, E., Remer, L. A., Kaufman, Y. J., Tanré, D., and Justice, C. O. (1994a), Atmospheric correction algorithm: spectral reflectances (MOD09), EOS MODIS Algorithm Technical Background Document, NASA Goddard Space Flight Center, Code 900, Greenbelt, Maryland 20771, USA, 30 pp.

Vermote, E., Tanré, D., Deuzé, J. L., Herman, M., and Morcrette, J. J. (1994b), Second stimulation of the satellite signal in the solar spectrum (6S), *6S User Guide Version 0*, 18 April, LOA-USTL, 59655 Villeneuve d'Ascq Cedex, France.